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- Getting Started
- Course Overview
- Schedule
- VM Access Guide
- AI Usage
- My Report
- Individual Projects



- Unit 2
- Unit 3
- Unit 4
- Unit 5
- Unit 6
- Unit 7

Group Capstone

- Groupings
- Career Resources

Pework

- Overview
- Lab
- Project
- Resources
- Career
- Report

Unit Tip: Your Ubuntu Course VM is **optional for this unit**. If you don't have access set up yet, you can follow the instructions to [Install Wireshark](#) on your local computer!

Warning to WINDOWS USERS - You will **not** be able to use your local machine for the Unit 1 [project](#). If you have not already, please be sure to [set up an Ubuntu VM](#) after this lab.

Lab #1: It Wasn't Me

Overview

Welcome to your first lab! Today we will join the Blue Team by carrying out the following scenario: You've been hired to the Security Operations Center of the company Boring Office to help track down the employee behind some nefarious activity. On April 19th, 2023 at 12:50PM an employee at Boring Office sent their deepest darkest secrets to everyone within the company. Or did they? This employee claims that someone else at the company is impersonating them, and your job is to hunt for who this rogue user is.

Goals

By the end of this lab you will be able to...

- Analyze a .pcap file to find a user's IP address
- Use DHCP logs to correlate an IP address to a host device
- Identify which user was logged in to a host device using account security logs

Note: There are no stretch goals for this lab.

Resources

- SMTP
- IP Address
- Network Layer
- DHCP

Lab Instructions

Exercise 0: Getting Started

For this lab, you will need:

A working installation of [Wireshark](#).

The three lab files:

1. [SMTP.pcap](#)
2. [DHCP.txt](#)
3. [Security_log.rtf](#)



ACCESSING WIRESHARK

- Option A:  [Download and install Wireshark](#) on your own device to analyze `.pcap` files.
 Which version should I download?
 - -
 -
 -
 -
-
- Option B: If you have access, use your *Course VM*  [\(access guide\)](#).

Note: There should be no difference between these two methods, but running on your local machine might help things be less laggy.

ACCESSING THE FILES

- Option A: If you're running Wireshark on your local machine:
 - Click the file names above to download them.
 - Download all files, and note where they download to (your Downloads folder by default)
 - Go to Wireshark, click on “*Open*” located under “Welcome to Wireshark”
 - Look in your user files for the Downloaded file, click on it and click *Open*
 - Now choose the file `smtp.pcap` and *Open* again
 - This will directly take you to the contents of the file!
- Option B: The files are on your provided *Course VM*  ([access guide](#)) at `~/lab_1` -- No downloads needed!

Exercise 1: Find the IP Address

Open up the .pcap file in Wireshark (see  *ACCESSING THE FILES* above)

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.10.1.4	10.10.1.1	DNS	76	Standard query 0x7956 A mail.patriots.in
2	0.034025	10.10.1.1	10.10.1.4	DNS	142	Standard query response 0x7956 A mail.patriots.in CNAME patriots.in A 74.53.140.1
3	0.036986	10.10.1.4	74.53.140.153	TCP	62	1470 → 25 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM
4	0.383936	74.53.140.153	10.10.1.4	TCP	62	25 → 1470 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM
5	0.383968	10.10.1.4	74.53.140.153	TCP	54	1470 → 25 [ACK] Seq=1 Ack=1 Win=65535 Len=0
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
8	1.073326	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=182 Ack=10 Win=5840 Len=0
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] SIZE 52428800 PIPEL
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
11	1.419021	74.53.140.153	10.10.1.4	SMTP	72	S: 334 VXNlcm5hbWU6
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: Z3VycGFydGFwQHhhdHJpb3RzLmlu
13	1.761484	74.53.140.153	10.10.1.4	SMTP	72	S: 334 UGFzc3dvcmQ6
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: cHVuamF1QDEyMw==
15	2.121738	74.53.140.153	10.10.1.4	SMTP	84	S: 235 Authentication succeeded
16	2.122354	10.10.1.4	74.53.140.153	SMTP	90	C: MAIL FROM: <gurpartap@patriots.in>
17	2.464705	74.53.140.153	10.10.1.4	SMTP	62	S: 250 OK
18	2.465190	10.10.1.4	74.53.140.153	SMTP	93	C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19	2.827648	74.53.140.153	10.10.1.4	SMTP	68	S: 250 Accepted
20	2.828143	10.10.1.4	74.53.140.153	SMTP	60	C: DATA
21	3.169619	74.53.140.153	10.10.1.4	SMTP	110	S: 354 Enter message, ending with "." on a line by itself
22	3.200683	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
23	3.200726	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes

Frame 1: 76 bytes on wire (608 bits), 76 bytes captured (608 bits)	0000	0010	0020	0030	0040
Ethernet II, Src: 00:e0:1c:3c:17:c2, Dst: 00:1f:33:d9:81:60	00 1f 33 d9 81 60 00 e0	1c 3c 17 c2 08 00 45 00	..3...<...E..		
Internet Protocol Version 4, Src: 10.10.1.4, Dst: 10.10.1.1	00 3e 25 0a 00 00 80 11	ff 8c 0a 0a 01 04 0a 0a	>%.....		
User Datagram Protocol, Src Port: 56166, Dst Port: 53	01 01 db 66 00 35 00 2a	8e 8d 79 56 01 00 00 01	...f.5*...yV...		
Domain Name System (query)	00 00 00 00 00 00 04 dd	61 69 6c 08 70 61 74 72	mail-patr		
	69 6f 74 73 02 69 6e 00	00 01 00 01	...ots.in....		

We should see 60 packets that we have captured. Combing through each packet to see if it has the information we want can be tiresome, especially if we have hundreds or even thousands of packets!. So instead, we'll use the **Display Filter** to sort out the unnecessary ones.

At the top where it says "Apply a display filter ..." let's go ahead and input `smtp` and hit enter.

No.	Time	Source	Destination	Protocol	Length	Info
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] SIZE 52428800 PIPEL
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
11	1.419021	74.53.140.153	10.10.1.4	SMTP	72	S: 334 VXNlcm5hbWU6
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: Z3VycGFydGFWQHBhdHJpb3RzLmlu
13	1.761484	74.53.140.153	10.10.1.4	SMTP	72	S: 334 UGFzc3dvcmQ6
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: cHVuamFiQDEyMw==
15	2.121738	74.53.140.153	10.10.1.4	SMTP	84	S: 235 Authentication succeeded
16	2.122354	10.10.1.4	74.53.140.153	SMTP	90	C: MAIL FROM: <gurpartap@patriots.in>
17	2.464705	74.53.140.153	10.10.1.4	SMTP	62	S: 250 OK
18	2.465190	10.10.1.4	74.53.140.153	SMTP	93	C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19	2.827648	74.53.140.153	10.10.1.4	SMTP	68	S: 250 Accepted
20	2.828143	10.10.1.4	74.53.140.153	SMTP	60	C: DATA
21	3.169619	74.53.140.153	10.10.1.4	SMTP	110	S: 354 Enter message, ending with "." on a line by itself
22	3.200683	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
23	3.200726	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
24	3.200744	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
25	3.200763	10.10.1.4	74.53.140.153	SMTP	1514	[TCP Window Full] C: DATA Fragment, 1460 bytes
26	3.203055	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmentation needed)
28	3.203563	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmentation needed)
29	3.204188	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmentation needed)
30	3.204574	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmentation needed)

> Frame 6: 235 bytes on wire (1880 bits), 235 bytes captured (1880 bits)
 > Ethernet II, Src: 00:1f:33:d9:81:60, Dst: 00:e0:1c:3c:17:c2
 > Internet Protocol Version 4, Src: 74.53.140.153, Dst: 10.10.1.4
 > Transmission Control Protocol, Src Port: 25, Dst Port: 1470, Seq: 1, Ack: 1, Len
 > Simple Mail Transfer Protocol

0000 00 e0 1c 3c 17 c2 00 1f 33 d9 81 60 08 00 45 60 ...<... 3...E'
 0010 00 dd 21 e2 40 00 32 06 43 fd 4a 35 8c 99 0a 0a ...!@.2. C:25...
 0020 01 04 00 19 05 be ae ec 61 b0 7e c4 53 b1 50 18 a~S-P-
 0030 16 d0 8a 44 00 00 32 32 30 2d 78 63 39 30 2e 77 ...D..22 0-xc90.w
 0040 65 62 73 69 74 65 77 65 6c 63 6f 6d 65 2e 63 6f ebsitewe lcome.co
 0050 6d 20 45 53 4d 54 50 20 45 78 69 6d 20 34 2e 36 m ESMTP Exim 4.6
 0060 39 20 23 31 20 4d 6f 6e 2c 20 30 35 20 4f 63 74 9 #1 Mon , 05 Oct
 0070 20 32 30 30 39 20 30 31 3a 30 35 3a 35 34 20 2d 2009 01 :05:54 -
 0080 30 35 30 30 20 0d 0a 32 32 30 2d 57 65 20 64 6f 0500 ..2 20-We do
 0090 20 6e 6f 74 20 61 75 74 68 6f 72 69 7a 65 20 74 not aut horize t
 00a0 68 65 20 75 73 65 20 6f 66 20 74 68 69 73 20 73 he use o f this s
 00b0 79 73 74 65 6d 20 74 6f 20 74 72 61 6e 73 70 6f ystem to transpo
 00c0 72 74 20 75 6e 73 6f 6c 69 63 69 74 65 64 2c 20 rt unsol icited,
 00d0 0d 0a 32 32 30 20 61 6e 64 2f 6f 72 20 62 75 6c ..220 an d/or bul
 00e0 6b 20 65 2d 6d 61 69 6c 2e 0d 0a k e-mail ...

We should see that this filtered out some packets, and the ones that remain all use the SMTP (Simple Mail Transfer Protocol). This protocol states how servers should go about sending and receiving electronic mail messages. Although we're certainly getting closer to what we want with this search filter, we still don't have what we really want to see - the SMTP packets from the rogue user.

So let's input `smtp contains "FROM"` into our **Display Filter**.

- Note: Spelling and format matters here, so make sure you copy the filter exactly!

The image shows a Wireshark packet capture of an SMTP message. The top pane displays a list of packets, with packet 16 selected. The packet details pane shows the structure of the packet: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Simple Mail Transfer Protocol. The packet bytes pane shows the raw data of the packet, including the email header and body.

No.	Time	Source	Destination	Protocol	Length	Info
16	2.122354	10.10.1.4	74.53.140.153	SMTP	90	C: MAIL FROM: <gurpartap@patriots.in>

Frame 16: 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface 0
 Ethernet II, Src: 00:e0:1c:3c:17:c2, Dst: 00:1f:33:d9:81:60
 Internet Protocol Version 4, Src: 10.10.1.4, Dst: 74.53.140.153
 Transmission Control Protocol, Src Port: 1470, Dst Port: 25, Seq: 70, Ack: 385, Win: 0, Len: 90
 Simple Mail Transfer Protocol

0000 00 1f 33 d9 81 60 00 e0 1c 3c 17 c2 08 00 45 00 ...3...<....E-
 0010 00 4c 25 33 40 00 00 06 f3 9c 0a 0a 01 04 4a 35 ...L%3g...J5
 0020 8c 99 05 be 00 19 7e c4 53 f6 ae ec 63 30 50 18S...c0P
 0030 fe 7f e1 a1 00 00 4d 41 49 4c 20 46 52 4f 4d 3aMA IL FROM:
 0040 20 3c 67 75 72 70 61 72 74 61 70 40 70 61 74 72 <gurpar tap@patr
 0050 69 6f 74 73 2e 69 6e 3e 0d 0a iots.in> ..

This filters out a single packet for us where the sender of the mail is specified. We should see in the info column that the mail is coming from the compromised user’s account, but what we really want to take note of is the source IP (Internet Protocol) address: **10.10.1.4**. This tells us where within the network the rogue user performed these actions, and by taking a look at our DHCP log, we can match this information up to the name of the host computer that the rogue user acted from as well.

Exercise 2: Correlate to the Host Computer

Now that we know which IP address to look out for (10.10.1.4), we want to use the DHCP log file to match it to the host device that was used. DHCP (Dynamic Host Configuration Protocol) is how a server goes about assigning IP addresses to hosts/devices on the network.

Open up the DHCP log, and notice that the structure of the topmost line is consistently formatted.

- `DHCP.txt` is a text file -- You can use any text editor!

Even though this is our first time reading a DHCP log, there are some parts we can still try to pick out. If you would like more in-depth information on each log event, this [article](#) gives more context for the corresponding Event ID (the left most column on each line).

Field	Description
ID	A DHCP server event ID code.
Date	The date on which this entry was logged on the DHCP server.
Time	The time at which this entry was logged on the DHCP server.
Description	A description of this DHCP server event.
IP Address	The IP address of the DHCP client.
Host Name	The host name of the DHCP client.
MAC Address	The media access control address used by the network adapter hardware of the client.

Follow the description sections of each line top to bottom to get a feel for what's happening.

This is likely still very confusing, but the good news is that we really don't need to understand a lot of this file. All we care about is the activity at the time of day when the sensitive messages were sent - **12:50PM**. So, if the emails were sent at 12:50PM, then the rogue user must have gained access *before* that. So we will want to specifically analyze the events that came just before that time.

Identify the six events that occurred right before 12:50PM (around 12:11PM).

Now that we've narrowed our search, we want to find the host that our IP address in question (10.10.1.4) is assigned to.

Locate the event at 12:11:27PM.

We should see that the event at 12:11:27PM does just what we were looking for and assigned the IP address used by the rogue user from our .pcap file to the host device **USER2**. Amazing! We now know which computer the rogue user acted from, and by taking a look at the security log from the host device, we should be able to determine which employee at the company was logged in during that time!

Exercise 3: Analyze the Security Log

Opening our system security file should show us the login access logs, and is how we can find the user logged in at the time that the sensitive emails were sent. We've determined that the host device in question is **USER2**, and looking at the security log file, we should see four events logged.

Open the Security Log file with any word-processing program. ( Microsoft Word,  WordPad,  Pages,  TextEdit, etc)

Take a moment to read through the logs. What do you notice?

The first two were a logon/logoff session on host computer USER1, and the corresponding user section states this was by Jane Doe. The last two, however, were a logon/logoff session on host computer USER2.

Find the corresponding user for these logs, and we should have our answer!

Not only is USER2 the host device we suspected, but if we take a look at the duration of time in which **John Doe** was logged in, this perfectly overlaps with the time in which the sensitive emails were sent! We've finally discovered our culprit, and John will certainly have some explaining to do.

 Congratulations, you've completed your first lab! 